CHALLENGE: DETECTING ALZHEIMER'S DISEASE AND FRONTOTEMPORAL DEMENTIA USING ELECTROENCEPHALOGRAMS

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ABSTRACT

In 2025, it was estimated that 771,939 people in Canada live with dementia, including Alzheimer's disease and Frontotemporal dementia. By 2030, this number is estimated to reach 1 million.

Early detection of these diseases is still not well developed, and current advances in Deep Learning offer promising results.

For this challenge, you will have the opportunity to work on a curated dataset made up of EEG recordings of healthy subjects and subjects diagnosed with Alzheimer's disease and Frontotemporal dementia.

1 ABOUT EEG

An electroencephalogram (EEG) is a test that measures brain electrical activity and can identify anomalies in brain waves linked to certain disorders. The test uses small, metal discs called electrodes that attach to the scalp. Brain cells communicate via electrical impulses, and this activity shows up as wavy lines on an EEG recording. Brain cells are active all the time, even during sleep.

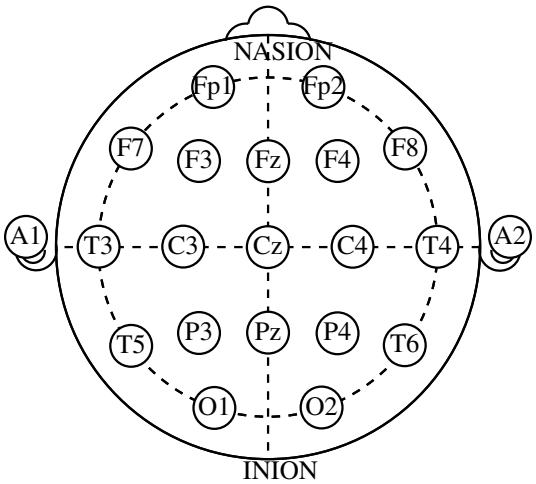

Figure 1: EEG Electrodes on a human skull

2 DATASET DESCRIPTION

This dataset contains the EEG resting state-closed eyes recordings from 88 subjects in total. A total of 36 of them were diagnosed with Alzheimer's disease (AD group), 23 were diagnosed with frontotemporal dementia (FTD group), and 29 were CN (healthy; CN is also written as C).

Recordings include the EEG signal from 19 scalp electrodes (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, and O2) and 2 reference electrodes, placed according to the

10–20 international system. The sampling rate was 500 Hz and the resolution was 10 uV/mm. Each recording lasted approximately 13.5 min for the AD group (min = 5.1, max = 21.3), 12 min for the FTD group (min = 7.9, max = 16.9), and 13.8 min for the CN group (min = 12.5, max = 16.5). In total, 485.5 min of AD, 276.5 min of FTD, and 402 min of CN recordings were collected and are included in the dataset. The timestep (ΔT) used is 4 seconds.

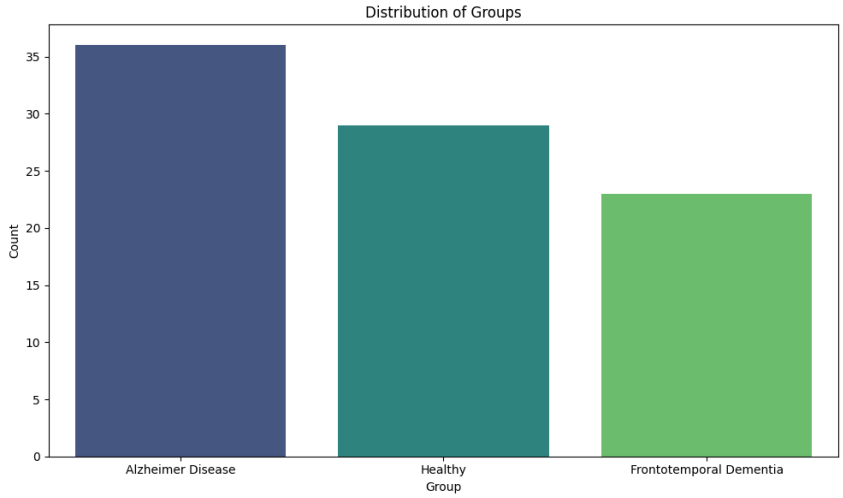

Figure 2: Distribution of classes in the dataset

You will also be given a CSV file mapping each subject with its condition, with A = Alzheimer's disease, C/CN = healthy, and F/FTD = frontotemporal dementia. Below is a preview of the file you will be given:

anonymized_id	label
2	F
3	A
4	A
10	C
13	F

Table 1: Label mapping for each subject

3 Feature extraction

As mentioned earlier, not all the EEG recordings have the same length. Hence, when downloading the data, you will notice samples with different shapes of the following format:

($num_{channels}$ = 19, $num_{timepoints}$)

Hence, you will not be able to concatenate all samples in one tensor/array. Fortunately, two options are presented to you:

- 1. Feature extraction: EEG can be downsized by extracting relevant features from the data which will be explained in further detail.
- 2. Padding/resizing: by converting all samples to the same size by either downsizing or upsizing.

3.1 Feature extraction

One of the most commonly extracted features for EEG classification tasks is the Relative Band Power (RBP) of the five frequency bands of interest of brain activity, namely:

- Delta: 0.5–4 Hz
- Theta: 4–8 Hz
- Alpha: 8–13 Hz
- Beta: 13–25 Hz
- Gamma: 25–45 Hz

Relative Power Band represents the ratio of power in a specific band to the total signal power. Another key feature extracted from the data is the Spectral Coherence Connectivity (SCC). It quantifies the synchronization of brain signals. It involves calculating the spectral coherence between each pair of signals, which measures the similarity of the frequency content between the two signals, and then averaging these values for each electrode:

$$SCC_x = \frac{1}{C} \sum_{y=1}^{C} \frac{|S_{xy}|}{\sqrt{S_{xx} * S_{yy}}} \tag{1}$$

To address the different lengths of the EEG recordings, we extract additional epochs from each participant using a sliding window approach. Specifically, the raw signals are divided into 30-second windows with a 15-second overlap. This standardizes the input size and increases the number of training samples. You will find below in the notebook two helper functions, *generate_dataset* and *extract_features*, to help you extract those features from these windows. Note that while the original EEG dataset used a 500 Hz sampling frequency, the helper functions downsample the data to 128 Hz for computational efficiency.

3.1.1 Subject-level evaluation

Because the sliding window approach extracts multiple 30-second epochs for a single participant, your model will generate multiple predictions per person. To evaluate the overall recording for one person, you must aggregate these epoch-level predictions to output a single final prediction per subject. You can achieve this using approaches such as:

- **Majority Voting**: Assign the patient the class most frequently predicted across all of their individual epochs.
- **Average Probability**: Calculate the average of the prediction probabilities across all epochs for a given patient, and classify them based on a threshold (e.g., 0.5).

Note: the dataset generation function needs an argument called *subjects*, which is the list of patients that are relevant to the task you'll choose (AD vs CN or FTD vs CN). To retrieve subjects, you can use the *filter_subjects* helper function, which will return the group subjects based on the task chosen. Make sure the subject identifiers match the EEG recordings in the data list!

3.2 Padding/Resizing

Instead of extracting specific features from the recording, you can also pad each sample to match the recording with the largest number of timestamps. Out of all the recordings in both training and testing sets, the one with the highest time points (in other words, the longest recording) is 640860. ATTENTION: this approach is computationally expensive and not the most efficient, but it guarantees exactitude of recordings.

On the other hand, you could try resizing it to a lower timestep. The sample with the lowest time points (i.e., the shortest recording) has 153050 time points. This approach is more computationally efficient. Again, helper functions to do both these operations are in the notebook.

4 Problem

Using the EEG recordings, there are two binary classification tasks you can choose:

1. AD vs CN: classify whether a patient has Alzheimer's disease or not
2. FTD vs CN: classify whether a patient has Frontotemporal dementia or not

5 EVALUATION

A non-disclosed scoring function will be used to evaluate your predictions on the test set, which will be released on Sunday at 9:30 a.m. We recommend monitoring Accuracy and F1-score during your internal validation. Participants will be evaluated on the following:

- a short presentation ($\sim$5–10 minutes) that clearly explains the problem, approach, results, and insights; a **slide deck is recommended**.
- a Jupyter notebook or GitHub repository with reproducible code.

Participants will be given access to a Google Drive folder with the following files:

- *example.npy*: the saved EEG recording for subject *example* saved as a npy file.
- *train_label_mapping.csv*: a CSV file mapping each subject with its condition (*C/CN = healthy, A = Alzheimer, F/FTD = Frontotemporal dementia*).

Final evaluation will use the following criteria:

1. **Methodology and Explorations (20 points):**
   - **Sound ML/DL Practice:** How appropriately your team handles preprocessing, feature construction, splits, and avoids leakage.
   - **Strategy and Justification:** How clearly your team defines and justifies the modeling strategy and methods chosen.
2. **Results and Performance (15 points):**
   - **Predictive Performance:** Quality of your model predictions based on the undisclosed scoring function.
   - **Validation Strategy:** Presentation of a structured pipeline and robustness checks.
   - **Generalization:** Reliability beyond training data and resistance to overfitting.
3. **Interpretability, Analysis, and Presentation Quality (15 points):**
   - **Interpretability:** Effort made to explain the model, its outputs, and the factors driving performance.
   - **Clarity:** Code readability, documentation, comments, reproducibility, and visualizations.

6 TRAINING DATA

All the training data can be downloaded from the following Google Drive folder link:

https://drive.google.com/drive/folders/1P4ai1Bq7GH3ZU8RtuwT1SseHErud4rlE?usp=sharing

All **test data** will be uploaded to the same location on Sunday at 9:30 am.

7 STARTER CODE

As mentioned earlier, a starting notebook (link is embedded just click) will be given to you with relevant helper functions for this challenge. Here is a snippet of the code:

```
# Example starter code
path = '/content/drive/MyDrive/example.npy' # Replace with your path
data = load_npy(path)

n_channels, time_points = data.shape
print(f'Number of channels: {n_channels}, Number of time_points: {
    time_points}')
```

```
# Visualize the sample data
fs = 500
duration = 5
max_samples = int(fs * duration)

subset_data = data[:5, :max_samples]
time = np.arange(max_samples) / fs

spacing = 100
scaling_factor = spacing / (np.std(subset_data) * 4)
scaled_data = subset_data * scaling_factor

# 3. Plotting
plt.figure(figsize=(12, 6))
offsets = np.arange(len(subset_data)) * spacing

for i in range(len(subset_data)):
    plt.plot(time, scaled_data[i] + offsets[i], linewidth=1, color='
    midnightblue')

# 4. Cleanup
plt.yticks(offsets, [f'CH {i+1}' for i in range(len(subset_data))])
plt.xlabel('Time (s)')
plt.title('Zoomed EEG View (First 5 Seconds)')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()

plt.show()
```

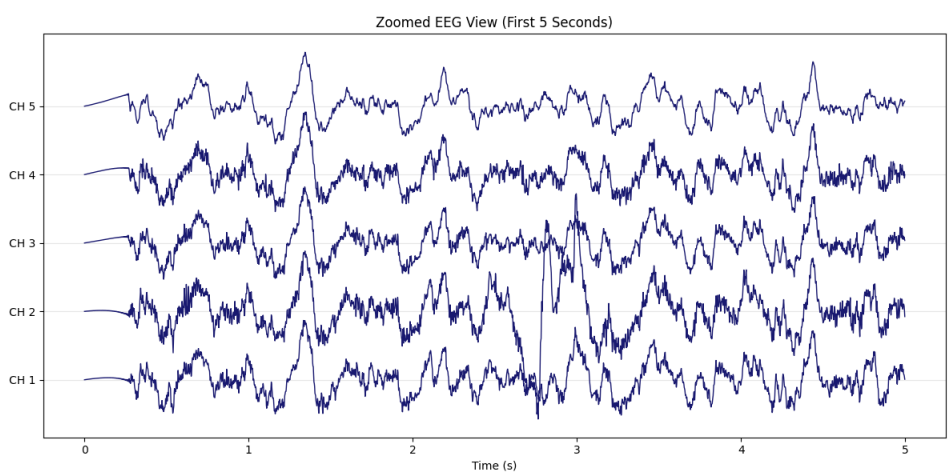

Figure 3: EEG plot

8 RESOURCES

The following is a list of resources that can be helpful. Using them is highly recommended as the challenge was inspired by these papers:

- *A Dataset of Scalp EEG Recordings of Alzheimer's Disease, Frontotemporal Dementia and Healthy Subjects from Routine EEG* [1]: This paper is the core of the hackathon. It explains how the EEG recordings were collected, how they processed the data, and what models they implemented.
- *DICE-Net: A Novel Convolution-Transformer Architecture for Alzheimer Detection in EEG Signals* [2]: This paper was written by the same authors. They introduce the DICE-Net, a powerful architecture specifically designed for Alzheimer's disease but also applicable to frontotemporal dementia.

9 DISCLAIMER

This challenge is not recommended for participants with limited ML experience and may be demanding. While high predictive performance is important, a sound methodology and clear insights are equally weighted in our evaluation.

The test set will be released at 9:30 a.m. on Sunday.

For submission, submit your notebook and slides to a GitHub repository and post them on Devpost. The devpost will be shared a bit after the start hackathon.

REFERENCES

[1] Andreas Miltiadous et al. "A Dataset of Scalp EEG Recordings of Alzheimer's Disease, Frontotemporal Dementia and Healthy Subjects from Routine EEG". In: *Data* 8.6 (2023). ISSN: 2306-5729. DOI: 10.3390/data8060095. URL: https://www.mdpi.com/2306-5729/8/6/95.

[2] Andreas Miltiadous et al. "DICE-Net: A Novel Convolution-Transformer Architecture for Alzheimer Detection in EEG Signals". In: *IEEE Access* 11 (2023), pp. 71840–71858. DOI: 10.1109/ACCESS.2023.3294618.